

Case Study 2

Creating database and tables.

```
MariaDB [(none)]> create database Projects;  
Query OK, 1 row affected (0.002 sec)
```

```
MariaDB [(none)]> use Projects;  
Database changed
```

```
MariaDB [Projects]> create table Employee(empNo varchar(10) primary key, fName  
varchar(20), lName varchar(20), address varchar(25), DOB date, sex varchar(10),  
position varchar(15), deptNo varchar(10));  
Query OK, 0 rows affected (0.028 sec)
```

```
MariaDB [Projects]> create table Department(deptNo varchar(10), deptName  
varchar(20), mgrEmpNo varchar(10), foreign key(mgrEmpNo) references  
Employee(empNo), primary key(deptNo));  
Query OK, 0 rows affected (0.025 sec)
```

```
MariaDB [Projects]> alter table employee add constraint fk_deptNo foreign  
key(deptNo) references Department(deptNo);  
Query OK, 0 rows affected (0.033 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [Projects]> create table Project(projNo varchar(10) primary key unique,  
projName varchar(20), deptNo varchar(10), foreign key(deptNo) references  
Department(deptNo));  
Query OK, 0 rows affected (0.008 sec)
```

```
MariaDB [Projects]> create table WorksOn(empNo varchar(10), projNo varchar(10),  
dateWorked date, hoursWorked int, foreign key(empNo) references Employee(empNo),  
foreign key(projNo) references Project(projNo), primary  
key(empNo,projNo,dateWorked));  
Query OK, 0 rows affected (0.016 sec)
```

```
describe employee; describe department; describe project; describe WorksOn;
```

```
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| empNo      | varchar(10)   | NO   | PRI | NULL    |       |  
| fName      | varchar(20)   | YES  |     | NULL    |       |  
| lName      | varchar(20)   | YES  |     | NULL    |       |  
| address    | varchar(25)   | YES  |     | NULL    |       |  
| DOB        | date          | YES  |     | NULL    |       |  
| sex        | varchar(10)   | YES  |     | NULL    |       |  
| position   | varchar(15)   | YES  |     | NULL    |       |  
| deptNo     | varchar(10)   | YES  | MUL | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
8 rows in set (0.028 sec)
```

```
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| deptNo     | varchar(10)   | NO   | PRI | NULL    |       |  
| deptName   | varchar(20)   | YES  |     | NULL    |       |  
| mgrEmpNo   | varchar(10)   | YES  | MUL | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.005 sec)
```

```
+-----+-----+-----+-----+-----+-----+
```

Field	Type	Null	Key	Default	Extra
projNo	varchar(10)	NO	PRI	NULL	
projName	varchar(20)	YES		NULL	
deptNo	varchar(10)	YES	MUL	NULL	

3 rows in set (0.006 sec)

Field	Type	Null	Key	Default	Extra
empNo	varchar(10)	NO	PRI	NULL	
projNo	varchar(10)	NO	PRI	NULL	
dateWorked	date	NO	PRI	NULL	
hoursWorked	int(11)	YES		NULL	

4 rows in set (0.008 sec)

7.24

```
MariaDB [Projects]> alter table employee add constraint sex_constraint
check(sex='M' or sex='F');
Query OK, 0 rows affected (0.036 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [Projects]> alter table employee add constraint post_constraint
check(position="Manager" or position="Team Leader" or position="Analyst" or
position="Software Developer");
Query OK, 0 rows affected (0.036 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [Projects]> alter table WorksOn add constraint check(hoursWorked between
0 and 40);
Query OK, 0 rows affected (0.035 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Inserting values

```
MariaDB [Projects]> INSERT INTO Department VALUES
-> ('D001','IT',NULL),
-> ('D002','Finance',NULL),
-> ('D003','Human Resources',NULL),
-> ('D004','Marketing',NULL),
-> ('D005','Operations',NULL),
-> ('D006','Research',NULL),
-> ('D007','Sales',NULL),
-> ('D008','Support',NULL),
-> ('D009','Engineering',NULL),
-> ('D010','Administration',NULL);
Query OK, 10 rows affected (0.009 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
MariaDB [Projects]> INSERT INTO Employee VALUES
-> ('E001','John','Mensah','Accra, Ghana','1980-05-
12','M','Manager','D001'),
-> ('E002','Ama','Boateng','Kumasi, Ghana','1985-03-22','F','Team
Leader','D002'),
-> ('E003','Kwame','Asante','Takoradi, Ghana','1990-07-
15','M','Analyst','D003'),
-> ('E004','Mary','Owusu','Cape Coast, Ghana','1988-09-10','F','Software
Developer','D004'),
-> ('E005','David','Arthur','Johannesburg, South Africa','1982-11-
30','M','Manager','D005'),
-> ('E006','Linda','Ofori','Mumbai, India','1991-01-
```

```

18','F','Analyst','D006'),
-> ('E007','Michael','Addo','Beijing, China','1987-06-25','M','Software
Developer','D007'),
-> ('E008','Sarah','Mensima','Sao Paulo, Brazil','1992-04-08','F','Team
Leader','D008'),
-> ('E009','Peter','Nyarko','Moscow, Russia','1984-08-
14','M','Manager','D009'),
-> ('E010','Grace','Antwi','Accra, Ghana','1993-12-03','F','Software
Developer','D010');
Query OK, 10 rows affected (0.011 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E001' WHERE deptNo='D001';
Query OK, 1 row affected (0.015 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E002' WHERE deptNo='D002';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E003' WHERE deptNo='D003';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E004' WHERE deptNo='D004';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E005' WHERE deptNo='D005';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E006' WHERE deptNo='D006';
Query OK, 1 row affected (0.002 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E007' WHERE deptNo='D007';
Query OK, 1 row affected (0.002 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E008' WHERE deptNo='D008';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E009' WHERE deptNo='D009';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [Projects]> UPDATE Department SET mgrEmpNo='E010' WHERE deptNo='D010';
Query OK, 1 row affected (0.010 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

MariaDB [Projects]> INSERT INTO Project VALUES
-> ('P001','Payroll System','D001'),
-> ('P002','Budget Analysis','D002'),
-> ('P003','Recruitment App','D003'),
-> ('P004','Ad Campaign','D004'),
-> ('P005','Inventory System','D005'),
-> ('P006','AI Research','D006'),
-> ('P007','Sales Portal','D007'),
-> ('P008','Help Desk','D008'),
-> ('P009','Automation Tool','D009'),
-> ('P010','Admin Dashboard','D010');
Query OK, 10 rows affected (0.009 sec)

```

Records: 10 Duplicates: 0 Warnings: 0

MariaDB [Projects]> INSERT INTO WorksOn VALUES

```
-> ('E001','P001','2013-01-15',35),
-> ('E002','P002','2013-02-20',30),
-> ('E003','P003','2013-03-10',25),
-> ('E004','P004','2013-04-05',40),
-> ('E005','P005','2014-05-12',38),
-> ('E006','P006','2014-06-18',32),
-> ('E007','P007','2015-07-22',28),
-> ('E008','P008','2015-08-14',20),
-> ('E009','P009','2016-09-30',36),
-> ('E010','P010','2016-10-11',24);
```

Query OK, 10 rows affected (0.011 sec)

Records: 10 Duplicates: 0 Warnings: 0

MariaDB [projects]> UPDATE Employee

```
-> SET position = 'Manager'
-> WHERE empNo IN ('E002','E006','E008');
```

Query OK, 3 rows affected (0.013 sec)

Rows matched: 3 Changed: 3 Warnings: 0

MariaDB [projects]>

MariaDB [projects]> UPDATE Department

```
-> SET mgrEmpNo = 'E002'
-> WHERE deptNo = 'D002';
```

Query OK, 0 rows affected (0.001 sec)

Rows matched: 1 Changed: 0 Warnings: 0

MariaDB [projects]>

MariaDB [projects]> UPDATE Department

```
-> SET mgrEmpNo = 'E006'
-> WHERE deptNo = 'D006';
```

Query OK, 0 rows affected (0.002 sec)

Rows matched: 1 Changed: 0 Warnings: 0

MariaDB [projects]>

MariaDB [projects]> UPDATE Department

```
-> SET mgrEmpNo = 'E008'
-> WHERE deptNo = 'D008';
```

Query OK, 0 rows affected (0.001 sec)

Rows matched: 1 Changed: 0 Warnings: 0

7.25

create view FemaleProjectManagers as select

p.projNo,p.projName,e.empNo,e.fName,e.lName,e.sex,e.position from project p join

-> department d on d.deptNo=p.deptNo

-> join (Select empNo, fName,lName, position, sex from employee where
sex="F" and position="Manager") e on e.empNo=d.mgrEmpNo

-> order by p.projNo;

Query OK, 0 rows affected (0.014 sec)

MariaDB [projects]> SELECT * FROM FemaleProjectManagers;

projNo	projName	empNo	fName	lName	sex	position
P002	Budget Analysis	E002	Ama	Boateng	F	Manager
P006	AI Research	E006	Linda	Ofori	F	Manager
P008	Help Desk	E008	Sarah	Mensima	F	Manager

3 rows in set (0.001 sec)

7.26

```

MariaDB [projects]> create view EmployeePorjects as select e.empNo,
e.fName,e.lName, p.projName, w.hoursWorked from employee e join
-> department d on d.deptNo=e.deptNo
-> join project p on p.deptNo=d.deptNo
-> join WorksOn w where w.projNo=p.projNo;
Query OK, 0 rows affected (0.014 sec)

```

```

MariaDB [projects]> select* from Employeeporjects;
+-----+-----+-----+-----+-----+
| empNo | fName | lName | projName | hoursWorked |
+-----+-----+-----+-----+-----+
| E001 | John | Mensah | Payroll System | 35 |
| E002 | Ama | Boateng | Budget Analysis | 30 |
| E003 | Kwame | Asante | Recruitment App | 25 |
| E004 | Mary | Owusu | Ad Campaign | 40 |
| E005 | David | Arthur | Inventory System | 38 |
| E006 | Linda | Ofori | AI Research | 32 |
| E007 | Michael | Addo | Sales Portal | 28 |
| E008 | Sarah | Mensima | Help Desk | 20 |
| E009 | Peter | Nyarko | Automation Tool | 36 |
| E010 | Grace | Antwi | Admin Dashboard | 24 |
+-----+-----+-----+-----+-----+
10 rows in set (0.003 sec)

```

7.27

```

MariaDB [projects]> CREATE VIEW EmpProject (empNo, projNo, totalHours)
-> AS
-> SELECT
->     w.empNo,
->     w.projNo,
->     SUM(w.hoursWorked) AS totalHours
-> FROM Employee e, Project p, WorksOn w
-> WHERE e.empNo = w.empNo
->     AND p.projNo = w.projNo
-> GROUP BY w.empNo, w.projNo;
Query OK, 0 rows affected (0.017 sec)

```

```

MariaDB [projects]> Select* from EmpProject;
+-----+-----+-----+
| empNo | projNo | totalHours |
+-----+-----+-----+
| E001 | P001 | 35 |
| E002 | P002 | 30 |
| E003 | P003 | 25 |
| E004 | P004 | 40 |
| E005 | P005 | 38 |
| E006 | P006 | 32 |
| E007 | P007 | 28 |
| E008 | P008 | 20 |
| E009 | P009 | 36 |
| E010 | P010 | 24 |
+-----+-----+-----+
10 rows in set (0.011 sec)

```

```

MariaDB [projects]> select projNo from EmpProject where projNo='SCCS';
+-----+
| projNo |
+-----+
| SCCS   |
+-----+
1 row in set (0.010 sec)

```

```

MariaDB [projects]> select count(projNo) from empProject where empNo='E1';
+-----+

```

```
| count(projNo) |
+-----+
|          1 |
+-----+
1 row in set (0.001 sec)
```

MariaDB [projects]> select empNo, totalHours from empproject group by empNo;

```
+-----+-----+
| empNo | totalHours |
+-----+-----+
| E001  |          35 |
| E002  |          30 |
| E003  |          25 |
| E004  |          40 |
| E005  |          38 |
| E006  |          32 |
| E007  |          28 |
| E008  |          20 |
| E009  |          36 |
| E010  |          24 |
| E1    |          21 |
+-----+-----+
```